

Supplementary information

A Cyclization and Docking Protocol for Cyclic Peptide-Protein Modelling using HADDOCK2.4

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Supplementary Methods

Another metric typically used for interface accuracy is the interface root-mean square deviation (i-RMSD), which is calculated on backbone atoms of both protein and peptide residues that are within a 10 Å radius from each other in the reference complex. However, i-RMSD fails to correctly evaluate acceptable cyclic peptide solutions as illustrated in Figure S1 in which the depicted model has the correct interface while only the flexible non-interacting peptide regions deviate from the crystal structure. This predicted complex scored high for f_{nat} (0.86) with a i-RMSD higher than the acceptable cutoff (2.43Å), demonstrating that the i-RMSD metric can be misleading for the quality of cyclic peptide binding modes. This is because i-RMSD focuses on the backbone conformation of the peptide, while the crucial determinant for cyclic peptide interactions are the correct contacts, mainly obtained via side chains. The f_{nat} metric is, therefore, ideal for reporting on the correct contacts and this is why it was chosen for the analysis in this study as well as in other studies concerning cyclic peptides²⁵.

Table S1. The Backbone dataset including 18 complexes composed of a receptor and a peptide cyclized through its N- and C- terminus. The length of the peptides varies from 6 to 14 residues and the corresponding *apo* structure of the receptor was found for all but one complex. The sequence of the peptides is shown with the additional disulfide bond highlighted in black in the respective cases, the termini are highlighted in boldface.

PDB ID <i>holo</i> <i>receptor</i>	Peptide Length	PDB ID <i>apo</i> <i>receptor</i> ^a	Resolution (Å)	Sequence ^b	Identifier
3wne	6	5jl4	1.70	PKIDNG	10.2210/pdb3WNE/pdb
3zgc	7	1u6d	2.20	GDEETGE	10.2210/pdb3ZGC/pdb
3av9	8	5jl4	1.70	SAKIDNLD	10.2210/pdb3AV9/pdb
3ava	8	5jl4	1.70	ALKIDNLD	10.2210/pdb3AVA/pdb
3avb	8	5jl4	1.85	SLKIDNLD	10.2210/pdb3AVB/pdb
3avf	8	5jl4	1.70	DLKIDNLD	10.2210/pdb3AVF/pdb
3avg	8	5jl4	1.70	ADKIDNLD	10.2210/pdb3AVG/pdb
3avh	8	5jl4	1.88	ARKIDNLD	10.2210/pdb3AVH/pdb
3avi	8	5jl4	1.70	SLKIDNMD	10.2210/pdb3AVI/pdb
3avj	8	5jl4	1.70	ALKIDNMD	10.2210/pdb3AVJ/pdb
3avk	8	5jl4	1.75	SLKIDNED	10.2210/pdb3AVK/pdb
3avm	8	5jl4	1.88	SRKIDNLD	10.2210/pdb3AVM/pdb
3avn	8	5jl4	2.10	SHKIDNLD	10.2210/pdb3AVN/pdb
5xn3	8	-	1.34	RGDINNIV	10.2210/pdb5XN3/pdb
1sfi	14	1tld	1.65	GR C TKSIPPI C FPD	10.2210/pdb1SFI/pdb
3p8f	14	4is5	2.00	GR C TKSIPPI C FPD	10.2210/pdb3P8F/pdb
4k1e	14	4kga	1.30	GF C QRSIPPI C FPD	10.2210/pdb4K1E/pdb
4kel	14	4kga	1.15	GF C QRSIPPI C FPN	10.2210/pdb4KEL/pdb

a. Absence of the apo receptor structure is indicated with a dash (" - ")

b. Highlighted in black is the additional disulfide bond found in the peptides of the respective complexes. In black bold face the connection of the termini is represented.

Table S2. Disulfide dataset including 12 complexes composed of a receptor and a peptide cyclized through a single disulfide bond. The length of the peptides varies from 6-14 residues and the corresponding apo structure of the receptor was found for 8 out of 12 complexes. As shown in the sequence of each peptide by a black highlight, only one disulfide bond is present in each peptide that cyclizes the sequence.

PDB ID <i>holo</i> <i>receptor</i>	Peptide Length	<i>apo</i> receptor PDB ^a	Resolution (Å)	Sequence ^b	Identifier
3wnf	6	5jl4	1.45	CKIDNC	10.2210/pdb3WNF/pdb
4ou3	6	4fkh	1.95	CNGRCG	10.2210/pdb4OU3/pdb
1smf	9	1tld	2.10	CTKSIPPEC	10.2210/pdb1SMF/pdb
3p72	11	1m0z	1.90	CTERMALHNL	10.2210/pdb3P72/pdb
2ck0	11	-	2.20	CKEWLSTAPCG	10.2210/pdb2CK0/pdb
5th2	12	-	1.84	CQFDQSTRRLKC	10.2210/pdb5TH2/pdb
5djc	13	5dj0	2.10	DCAWHLGELVWCT	10.2210/pdb5DJC/pdb
4ib5	13	3q9x	2.20	GCRLYGFKIHGCG	10.2210/pdb4IB5/pdb
5h5q	13	2obi	1.10	CRVDLQGWRR	10.2210/pdb5H5Q/pdb
5eoc	13	-	1.98	CQLINTNGSWHIC	10.2210/pdb5EOC/pdb
5wxr	14	4dva	1.75	GAC	10.2210/pdb5WXR/pdb
4m1d	14	-	1.80	CRIHIGPGRAFYTC	10.2210/pdb4M1D/pdb

a. Absence of the apo receptor structure is indicated with a dash ("-")

b. Highlighted in black is the disulfide bond that cyclizes the peptide.

Table S3. Parameters changed in the run.cns file for Step 2 of the cyclization protocol.
The indicated values are the optimized settings for Step 2 of cyclization.

Cyclization Protocol: Step 2				
Description	Keyword Changes			
Number of structures to dock	structures_0=400	structures_1=400	waterrefine=400	anastruc_1=400
Electrostatics	elecflag_0=false	elecflag_1=false		
Explicit solvent refinement	solvshell=true			
Extra flexibility ^a	nfle_1=1	start_fle_1_1="1"	end_fle_1_1="14"	
it0	crossdock=false	randorien=false	rigidmini=false	ntrials=1
it1	initiosteps=2000	cool1_steps=2000	cool2_steps=4000	cool3_steps=4000
Clustering	clust_meth="RMSD"	clust_cutoff=2.5		

a. These parameters correspond to a cyclic peptide being defined as molecule 1 and consisting of 14 residues.

Table S4. Parameters changed in the run.cns file for Step 3 of the cyclization protocol.
Highlighted in black are the additional settings which are unique for Step 3 and not found in Step 2. The indicated values are the optimized settings for Step 3 of cyclization.

Cyclization Protocol: Step 3				
Description	Keyword Changes			
Number of structures to dock	structures_0=400	structures_1=400	waterrefine=400	anastruc_1=400
Electrostatics	elecflag_0=false	elecflag_1=false		
Explicit solvent refinement	solvshell=true	watersteps=5000		
Extra flexibility ^a	nfle_1=1	start_fle_1_1="1"	end_fle_1_1="14"	
it0	crossdock=false	randorien=false	rigidmini=false	ntrials=1
it1	initiosteps=2000	cool1_steps=2000	cool2_steps=4000	cool3_steps=4000
Clustering	clust_meth="RMSD"	clust_cutoff=2.5		
Random seed ^b	iniseed=3			
Unambiguous restraints	unamb lastit=1			
Cyclic peptide setting ^c	cyclicpept_mol1=true			

a. These parameters correspond to a cyclic peptide being defined as molecule 1 and consisting of 14 residues.

b. The value used as a random seed is different for every repetition of Step 3. In this example the random seed used is 3.

c. This parameter corresponds to a cyclic peptide being defined as molecule 1.

Table S5. Parameters changed in **generate.inp** file (in the protocols directory of the run) for Step 3 of the cyclization protocol. Indicated values are the optimized settings for Step 3 of cyclization.

Cyclization Protocol: Step 3	
Description	Keyword Changes
Minimum distance between C- and N-termini to create hydrogen bond	cyclicpept_dist=3.5
Minimum distance between cysteines to create a disulfide bond	disulphide_dist=4.0

Table S6. Parameters changed in the **run.cns** file for all nine docking protocols tested in this study. The indicated values are the optimized settings for the docking runs.

Docking Protocol				
Description	Keyword Changes			
it1	initiosteps=2000	cool1_steps=2000	cool2_steps=4000	cool3_steps=4000
Number of structures to dock	structures_0=5000	structures_1=400	waterrefine=400	anastruc_1=400
Cyclic peptide setting ^a	cyclicpept_mol2=true			
Clustering	clust_meth="RMSD"	clust_cutoff=5		
Hydrogen bond restraints	hbonds_on=true			

a. This parameter corresponds to a cyclic peptide being defined as molecule 2.

Table S7. RMSD analysis of the Backbone peptide conformational ensemble. Reported are the number of peptides within a given (high, medium and acceptable) quality. Peptides were defined as high quality if their RMSD (in relation to the x-ray structure) was lower than 1.5 Å, medium if RMSD lower than 2.0 Å and acceptable if their RMSD was lower than 2.5 Å. Complexes are sorted according to peptide's sequence length (from short to long peptides).

Structure ensemble	Quality	PDB ID																		Average
		3wne	3zgc	3av9	3ava	3avb	3avf	3avg	3avh	3avi	3avj	3avk	3avm	3avn	5xn3	4kel	1sfi	3p8f	4k1e	
5	High	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Medium	1	2	2	0	0	0	1	1	1	0	0	0	1	1	0	0	0	0	1
	Acceptable	1	2	2	1	1	1	1	1	1	0	0	0	1	1	0	0	1	0	1
	Total	2	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
30	High	3	1	0	0	0	1	2	1	0	2	0	0	0	1	0	0	0	0	1
	Medium	3	8	3	0	3	1	4	3	2	3	2	0	3	3	0	0	0	1	2
	Acceptable	4	10	6	2	5	7	6	6	7	10	4	0	4	3	0	0	3	1	4
	Total	8	21	30	30	29	30	30	30	30	30	30	30	30	23	30	30	30	30	28
40	High	4	2	0	0	0	1	4	2	0	2	0	0	0	1	0	0	0	0	1
	Medium	4	10	3	0	3	2	6	4	2	5	4	0	3	4	0	0	0	1	3
	Acceptable	5	12	8	2	7	8	8	8	10	14	6	1	4	4	0	0	3	3	6
	Total	10	26	39	40	39	40	40	40	40	40	40	40	40	32	40	40	40	40	37
50	High	5	3	1	0	1	1	4	2	0	3	0	1	0	2	0	0	0	0	1
	Medium	5	11	4	1	4	3	7	5	2	6	5	1	3	5	0	0	1	1	4
	Acceptable	6	13	10	3	8	10	9	9	10	17	7	2	7	6	1	1	4	3	7
	Total	12	33	49	50	49	50	50	50	50	50	50	50	50	42	50	50	50	50	46
60	High	6	5	1	0	1	1	4	3	0	3	0	1	0	2	0	0	0	0	2
	Medium	7	13	4	1	5	3	9	7	4	8	6	2	3	6	0	0	1	1	4
	Acceptable	7	15	12	3	9	11	11	11	13	20	8	3	8	8	2	1	5	4	8
	Total	14	41	59	60	59	60	60	60	60	60	60	60	60	49	60	60	60	60	56

Table S8. RMSD analysis of the Disulfide peptide conformational ensemble. Reported are the number of peptides within a given (high, medium and acceptable) quality. Peptides were defined as high quality if their RMSD (in relation to the x-ray structure) was lower than 1.5 Å, medium if RMSD lower than 2.0 Å and acceptable if their RMSD was lower than 2.5 Å. Complexes are sorted according to peptide's sequence length (from short to long peptides) and residues that are not part of the cyclic region have been excluded from the RMSD calculation.

Structure ensemble	Quality	PDB ID												
		3wnf	4ou3	1smf	3p72	2ck0	5th2	5djc	4ib5	5h5q	5eoc	5wxr	4m1d	Average
5	High	0	2	0	0	0	0	0	0	0	0	0	0	0
	Medium	2	3	0	0	1	0	0	0	0	0	0	0	1
	Acceptable	4	4	1	0	1	0	0	0	0	0	0	0	1
	Total	5	4	5	5	5	5	2	5	1	5	5	4	4
30	High	2	3	0	0	0	0	0	0	0	0	0	0	0
	Medium	8	6	0	0	1	0	0	0	0	0	0	0	1
	Acceptable	15	10	2	0	5	0	0	0	0	0	1	0	3
	Total	23	10	27	30	30	26	4	19	8	30	30	17	21
40	High	3	3	0	0	0	0	0	0	0	0	0	0	1
	Medium	11	7	0	0	1	0	0	0	0	0	0	0	2
	Acceptable	18	12	3	0	7	0	0	1	0	0	1	0	4
	Total	30	13	36	40	40	33	5	24	11	40	40	19	28
50	High	4	3	0	0	0	0	0	0	0	0	0	0	1
	Medium	14	10	0	0	1	0	0	0	0	0	0	0	2
	Acceptable	23	15	3	0	9	0	0	1	0	0	1	0	4
	Total	37	16	45	50	50	41	8	30	14	50	50	25	35
60	High	5	3	0	0	0	0	0	0	0	0	0	0	1
	Medium	15	12	0	0	1	0	0	0	0	0	0	0	2
	Acceptable	26	17	3	0	10	0	0	2	0	0	1	0	5
	Total	43	19	54	60	60	51	9	35	16	60	60	30	41

Table S9. RMSD analysis of the Backbone peptide conformational ensemble. Reported are the best, worst and median RMSD values of every complex in all the different ensemble sizes. Complexes are sorted according to peptide's sequence length (from short to long peptides).

Backbone Dataset															
	5-structures RMSD			30-structures RMSD			40-structures RMSD			50-structures RMSD			60-structures RMSD		
	Best	Worst	Median	Best	Worst	Median	Best	Worst	Median	Best	Worst	Median	Best	Worst	Median
3wne	0.65	3.18	1.92	0.65	3.61	2.75	0.65	3.61	2.75	0.65	3.61	2.74	0.65	3.61	2.74
3zgc	1.35	3.67	2.81	1.35	4.39	2.78	1.06	4.39	2.79	1.06	4.39	2.86	1.06	4.39	2.88
3av9	1.71	4.47	2.67	1.71	4.47	3.18	1.71	4.92	3.23	1.14	4.92	3.23	1.14	4.92	3.23
3ava	2.16	4.55	3.82	2.01	5.16	3.53	2.01	5.16	3.39	1.65	5.16	3.4	1.65	5.16	3.4
3avb	2.12	4.46	3.34	1.62	4.48	3.34	1.62	4.48	3.2	1.44	4.52	3.22	1.44	4.54	3.23
3avf	2.42	4.69	3.77	1.34	4.69	3.26	1.34	4.95	3.26	1.34	4.95	3.33	1.34	5.3	3.37
3avg	1.87	3.94	3.47	0.72	4.6	3.1	0.72	4.6	3.09	0.72	4.6	3.11	0.72	4.6	3.12
3avh	1.63	5.15	3.54	1.24	5.3	3.3	1.24	5.3	3.3	1.24	5.3	3.29	1.24	5.3	3.3
3avi	1.98	4.7	3.68	1.95	4.77	3.35	1.95	4.77	3.45	1.95	4.77	3.45	1.73	4.86	3.52
3avj	2.55	4.88	2.82	1.14	4.88	2.8	1.14	4.88	2.8	1.14	4.88	2.8	1.14	4.88	2.8
3avk	2.56	4.64	3.83	1.67	4.7	3.42	1.67	4.7	3.47	1.67	4.7	3.42	1.67	4.7	3.42
3avm	2.57	4.2	3.7	2.57	5.16	3.55	2.49	5.16	3.55	1.5	5.16	3.54	1.5	5.16	3.55
3avn	1.67	4.42	3.18	1.67	5.13	3.39	1.67	5.13	3.25	1.67	5.13	3.25	1.67	5.29	3.38
5xn3	1.91	4.48	2.54	1.32	4.54	3.27	1.32	4.54	3.2	1.31	4.72	3.05	1.31	4.88	2.96
1sfi	2.96	4.57	3.9	2.73	5.19	3.89	2.73	5.63	3.94	2.15	5.63	3.94	2.15	5.63	3.91
4kel	2.63	4.06	3.82	2.54	5.23	4.04	2.54	5.48	4.0	2.33	5.48	3.91	2.33	5.48	3.91
3p8f	2.27	5.2	4.09	2.19	5.45	4.02	2.19	5.45	3.96	1.94	5.45	3.87	1.94	5.45	3.84
4k1e	2.68	4.44	3.76	1.83	5.0	4.01	1.83	5.11	4.01	1.83	5.26	3.99	1.83	5.26	4.04
Average	2.09	4.43	3.37	1.68	4.82	3.39	1.66	4.90	3.37	1.49	4.92	3.36	1.47	4.97	3.37

Table S10. RMSD analysis of the Disulfide peptide conformational ensemble. Reported are the best, worst and median RMSD values of every complex in all the different ensemble sizes. Complexes are sorted according to peptide's sequence length (from short to long peptides). Residues that are not part of the cyclic region have been excluded from the RMSD calculation.

Disulfide Dataset															
	5-structures RMSD			30-structures RMSD			40-structures RMSD			50-structures RMSD			60-structures RMSD		
	Best	Worst	Median	Best	Worst	Median	Best	Worst	Median	Best	Worst	Median	Best	Worst	Median
3wnf	1.61	3.3	2.31	1.02	3.34	2.28	1.02	3.53	2.29	0.96	3.85	2.31	0.58	3.85	2.32
4ou3	1.01	2.28	1.48	1.01	2.29	1.88	1.01	2.6	1.88	1.01	2.6	1.86	1.01	2.6	1.84
1smf	2.24	3.88	2.78	2.24	4.62	3.32	2.24	4.62	3.21	2.24	4.62	3.23	2.24	4.62	3.23
3p72	3.67	5.12	4.22	3.29	5.26	4.58	3.29	5.26	4.51	3.29	5.5	4.59	3.29	5.59	4.61
2ck0	1.76	4.13	3.27	1.76	4.8	3.32	1.76	4.98	3.36	1.76	4.98	3.36	1.76	4.98	3.38
5th2	4.97	6.01	5.56	4.71	6.06	5.61	4.71	6.06	5.67	4.71	6.06	5.57	4.71	6.45	5.57
5djc	4.85	5.5	5.2	4.85	5.7	5.52	4.85	5.92	5.53	4.33	6.33	5.52	4.33	6.33	5.53
4ib5	3.32	6.54	5.0	2.84	6.54	4.95	2.4	6.54	4.98	2.4	6.54	4.92	2.4	6.54	4.88
5h5q	5.57	5.57	5.57	2.72	5.57	4.00	2.72	6.03	4.01	2.72	6.04	3.97	2.72	7.03	3.97
5eoc	6.23	6.59	6.32	5.06	7.08	6.26	4.79	7.08	6.26	4.79	7.08	6.26	4.79	7.08	6.33
5wxr	2.56	4.1	3.77	2.4	4.81	3.75	2.4	4.81	3.75	2.4	4.81	3.63	2.4	5.46	3.62
4m1d	4.06	6.13	4.93	4.06	7.59	5.31	4.06	7.59	5.31	4.05	7.62	5.31	4.0	8.0	5.34
Average	3.49	4.93	4.20	3.00	5.31	4.23	2.94	5.42	4.23	2.86	5.58	4.21	2.84	5.73	4.22

Table S11. Analysis of 50STR_COMB generated models. Reported are the rank of the first acceptable model (according to f_{nat} and CAPRI criteria) and its f_{nat} and i-RMSD values respectively. Also, the rank of the best f_{nat} value is reported along with the i-RMSD and f_{nat} value for each specific model. Finally, reported is the total number of high, medium and acceptable models in every complex docking run according to their f_{nat} value and the CAPRI criteria. Complexes in the table are sorted according to the sequence length of their peptide (from short to long).

50STR_COMB									
	1 st acceptable model			Best f_{nat} model			Total number of models		
	Rank	i-RMSD	F_{nat}	Rank	i-RMSD	F_{nat}	High	Medium	Acceptable
3wne	1	1.53	0.91	7	1.2	0.96	32	72	222
3wnf	1	1.06	0.77	40	0.59	0.90	22	214	152
4ou3	1	1.77	0.45	4	0.61	0.87	32	117	194
3zgc	1	1.90	0.49	30	0.82	0.70	0	18	198
3av9	1	3.02	0.21	17	1.34	0.89	7	84	227
3avi	1	2.69	0.45	19	1.07	0.90	7	28	306
3avb	1	2.43	0.82	24	1.33	0.89	13	42	221
3avh	1	2.98	0.27	8	2.16	0.89	14	46	202
3avn	1	2.93	0.35	6	2.53	0.81	1	41	183
5xn3	1	2.91	0.29	172	1.62	0.64	0	23	272
3avf	1	2.45	0.30	21	2.07	0.89	9	83	184
3avk	1	1.98	0.82	43	1.90	0.86	7	43	157
3avm	1	1.92	0.56	14	1.24	0.85	1	29	240
3avg	1	2.99	0.27	8	1.86	0.96	8	54	199
3avj	1	2.69	0.34	10	1.14	0.84	1	28	259
3ava	1	3.00	0.32	313	2.23	0.88	2	45	283
1smf	1	1.29	0.71	50	0.96	0.89	8	137	189
3p72	1	3.07	0.27	185	2.01	0.49	0	0	96
2ck0	3	2.55	0.65	52	1.97	0.71	0	19	159
5th2	1	4.03	0.34	105	1.93	0.57	0	2	121
5djc	1	2.45	0.60	1	2.45	0.60	0	2	150
4ib5	3	4.01	0.26	95	3.32	0.54	0	7	127
5h5q	1	5.16	0.33	214	2.60	0.46	0	0	146
5eoc	6	5.88	0.20	25	3.74	0.46	0	0	70
5wxr	2	1.85	0.71	12	1.75	0.77	0	61	71
4k1e	1	2.81	0.41	41	1.65	0.68	0	12	125
4m1d	2	5.00	0.31	177	3.05	0.77	0	17	116
3p8f	2	1.89	0.53	19	1.45	0.66	0	11	93
1sfi	2	2.49	0.63	4	1.24	0.74	0	19	120
4kel	1	4.84	0.29	26	1.99	0.75	0	10	105

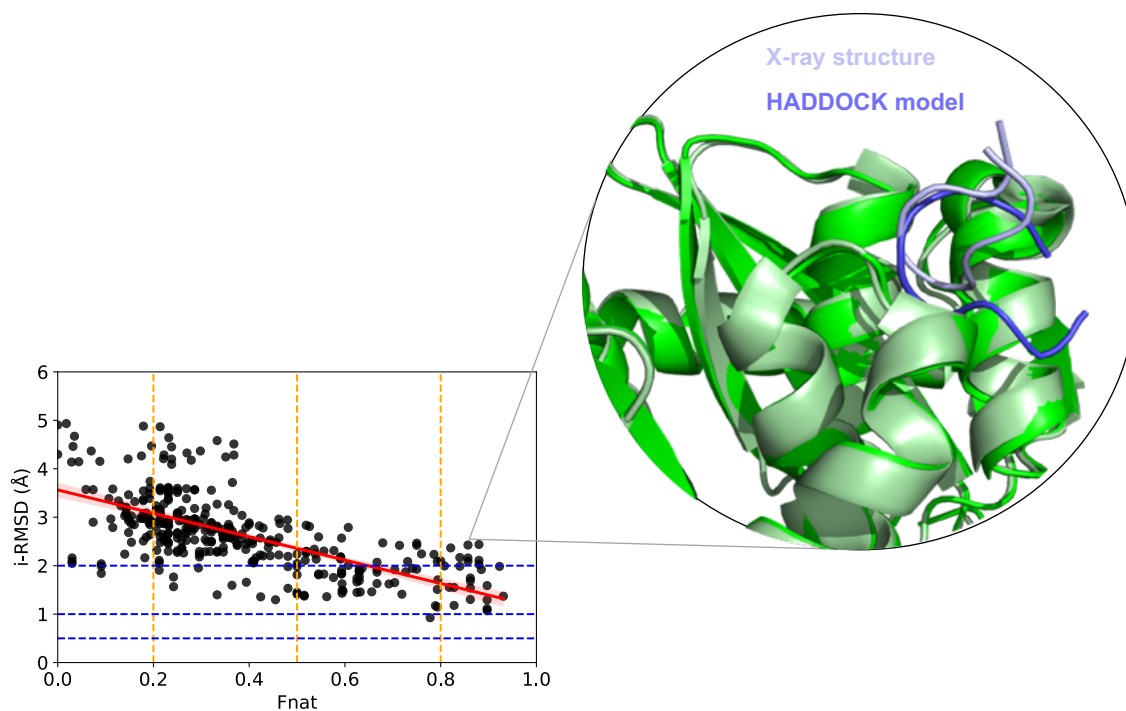


Figure S1. Scatter plot of i-RMSD and f_{nat} of top20 models for all complexes of the Backbone dataset, which were generated using the same protocol. 360 data points were used to create the scatter plot. Illustrated in the circle is 3av9 complex model that scored high f_{nat} (0.86) but also high i-RMSD (2.43).

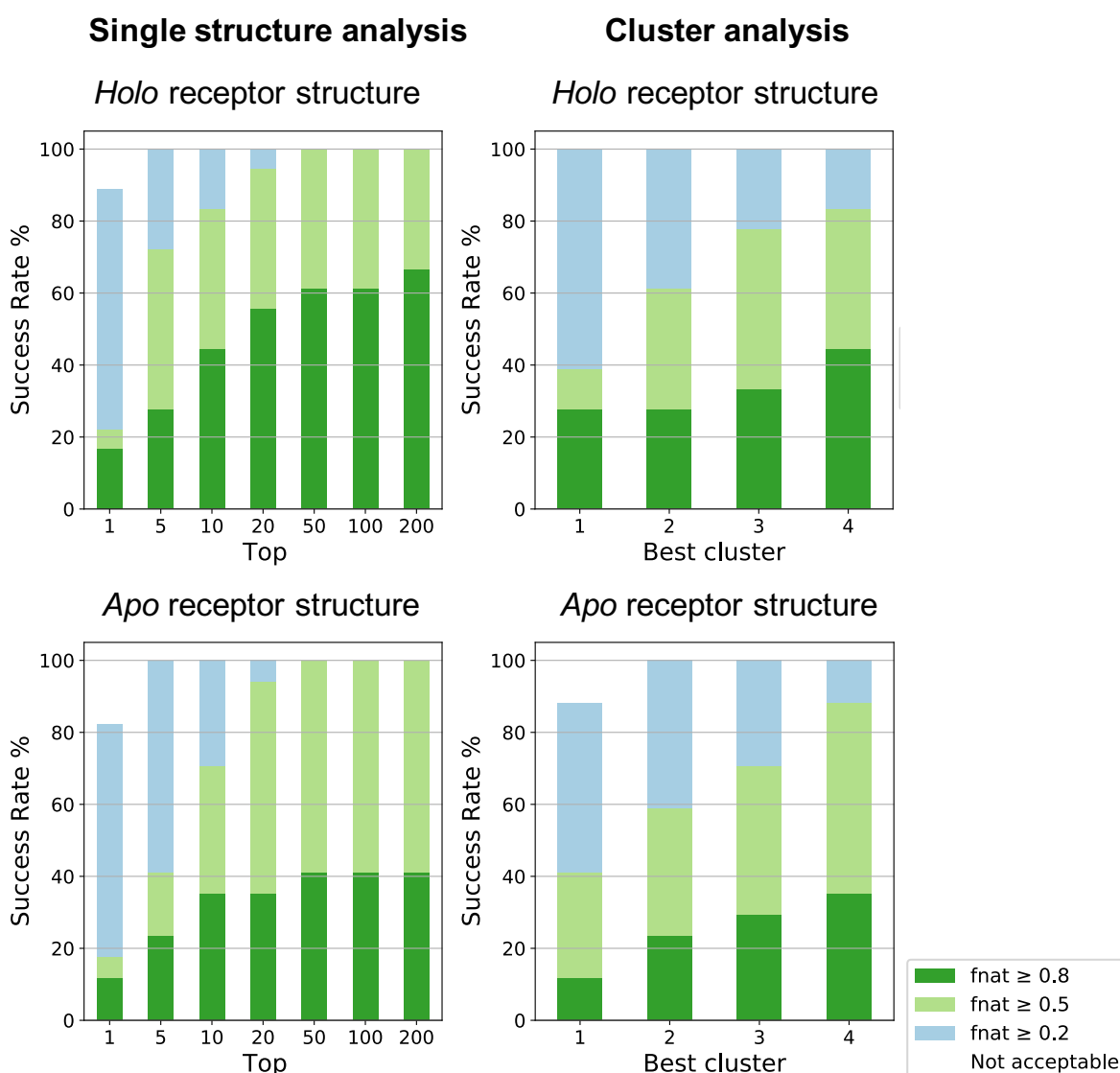


Figure S2. F_{nat} success rate (%) plots of the optimized experiment (50STR_COMB) for cyclic peptide using single structure or cluster analysis. Plotted on the left are the success rates of the different tops (top 1, top 5, top 10 etc.) of the single structure analysis. On the right, accumulatively plotted are the success rates of the 4 best clusters, according to the HADDOCK score of itw. Color coding (from blue to green) indicates the quality of the models (from acceptable to high) according to CAPRI criteria.

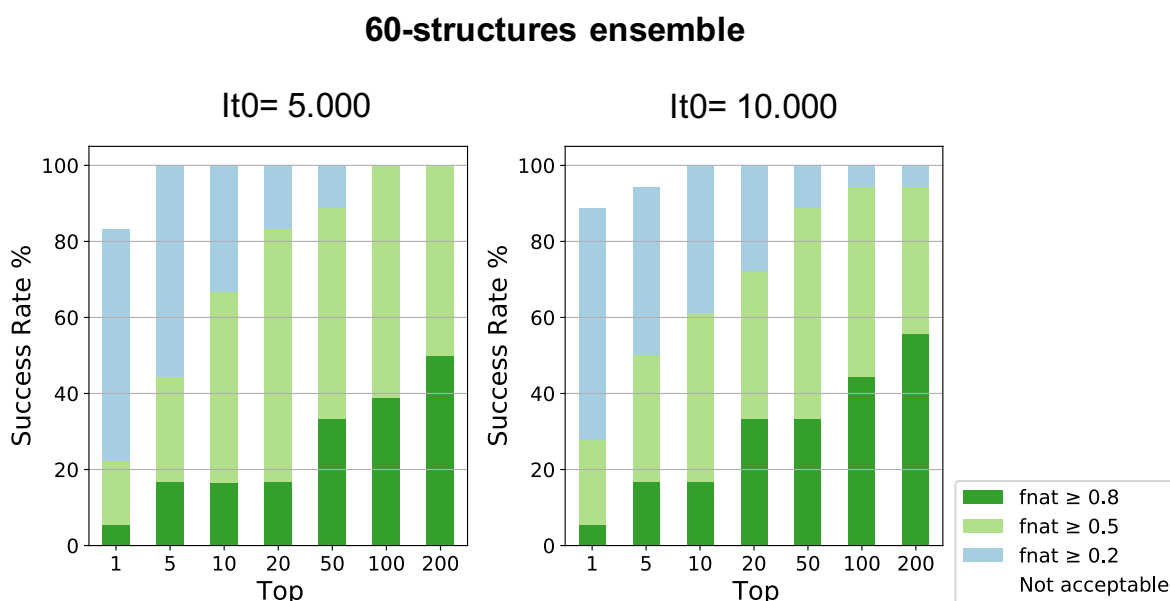


Figure S3. F_{nat} success rate (%) plots of experiments using a 60-structures ensemble and generating 5.000 or 10.000 models in it0. Plotted are the success rates of the Backbone dataset (n=18 complexes). Color coding (from blue to green) indicates model quality (acceptable to high), according to CAPRI criteria.

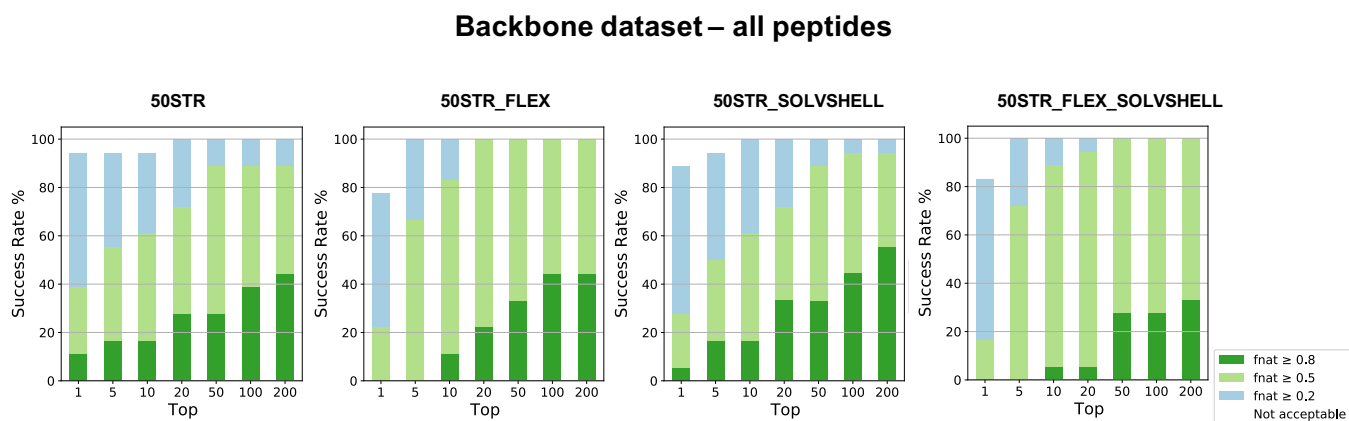


Figure S4. F_{nat} success-rate plots of 50STR_FLEX, 50STR_SOLVSHELL and 50STR docking protocols on the Backbone dataset. Plotted are the success rates of all Backbone complexes for each tested protocol, the color coding (from blue to green) indicates the quality of the models (from acceptable to high) according to CAPRI criteria.

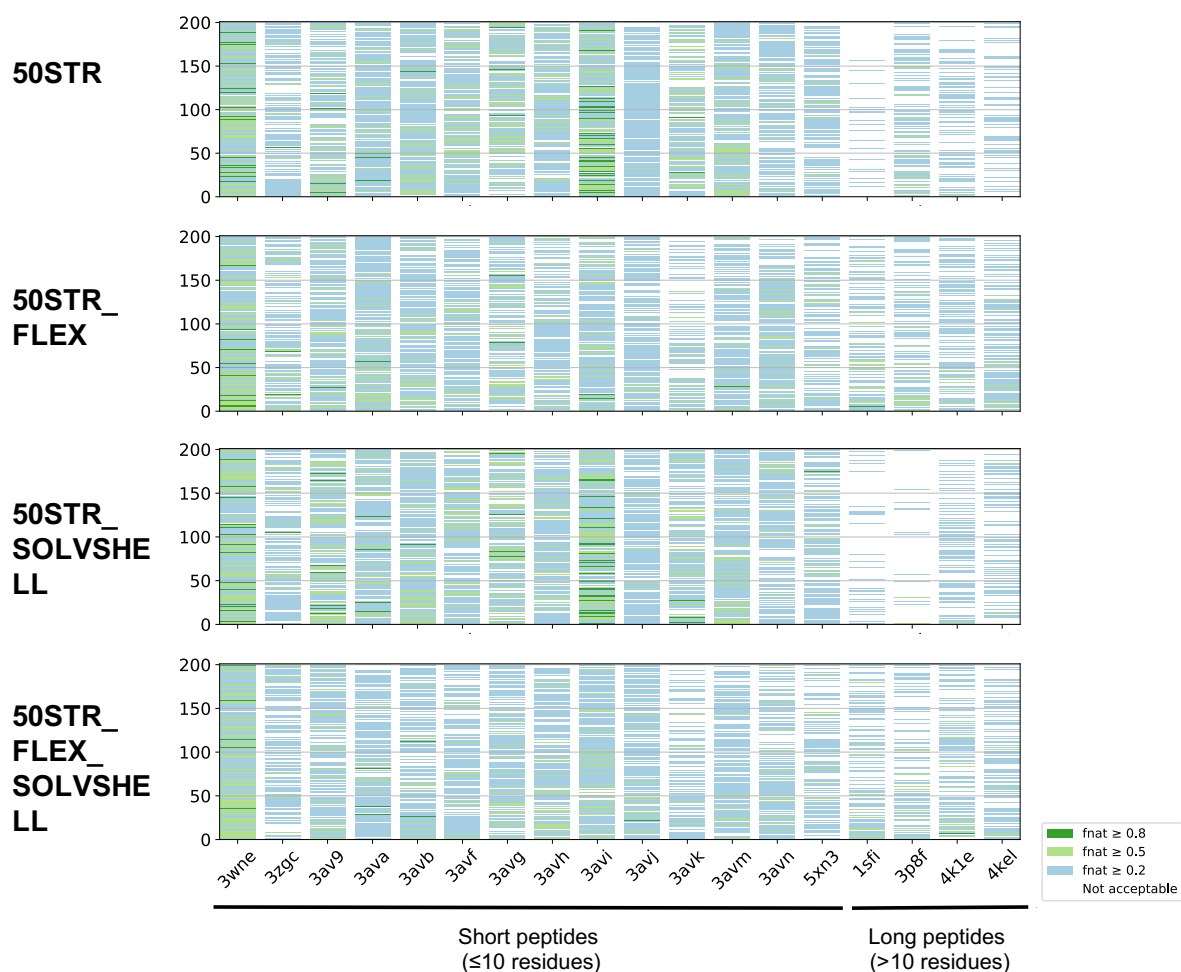


Figure S5. Comparison of performance of 50STR, 50STR_FLEX, 50STR_SOLVSHELL and 50STR_FLEX_SOLVSHELL protocols. Each column corresponds to one complex of the Backbone dataset and the y-axis reflects the ranking of the models according to the itw HADDOCK score in every protocol. The color of each model indicates its quality (from acceptable to high) according to the CAPRI criteria. Complexes are sorted in the x-axis according to their sequence length (from shorter to longer),

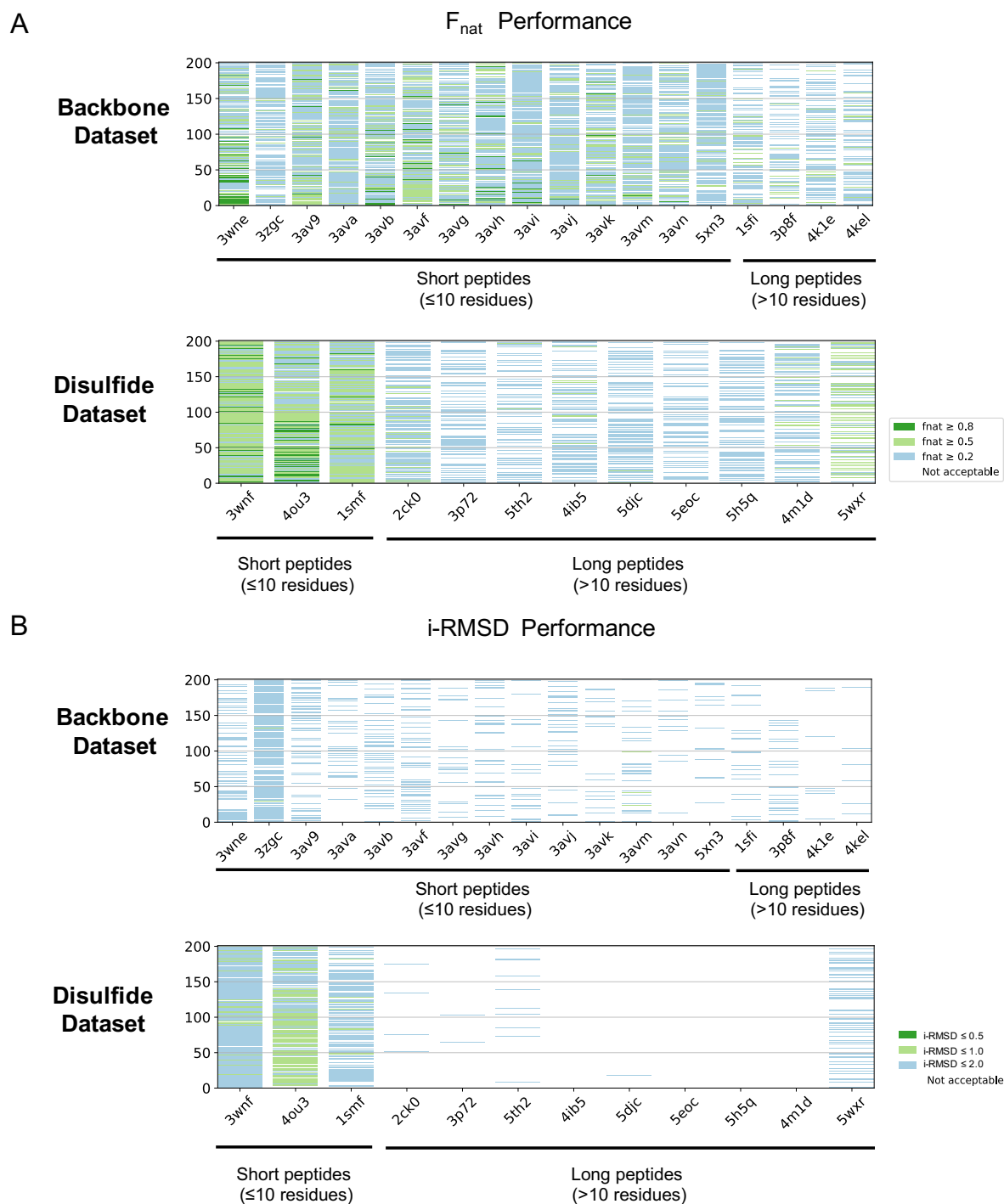


Figure S6. Comparison of performance of 50STR_COMB experiment in the Backbone and Disulfide dataset. Each column corresponds to one complex of the dataset and the y-axis reflects the ranking of the models according to the itw HADDOCK score in the protocol. Models are colored according to their f_{nat} score (**A**) or according to their i-RMSD (**B**). The color of each model indicates its quality (from acceptable to high) according to the CAPRI criteria. Complexes are sorted in the x-axis according to their sequence length (from shorter to longer).

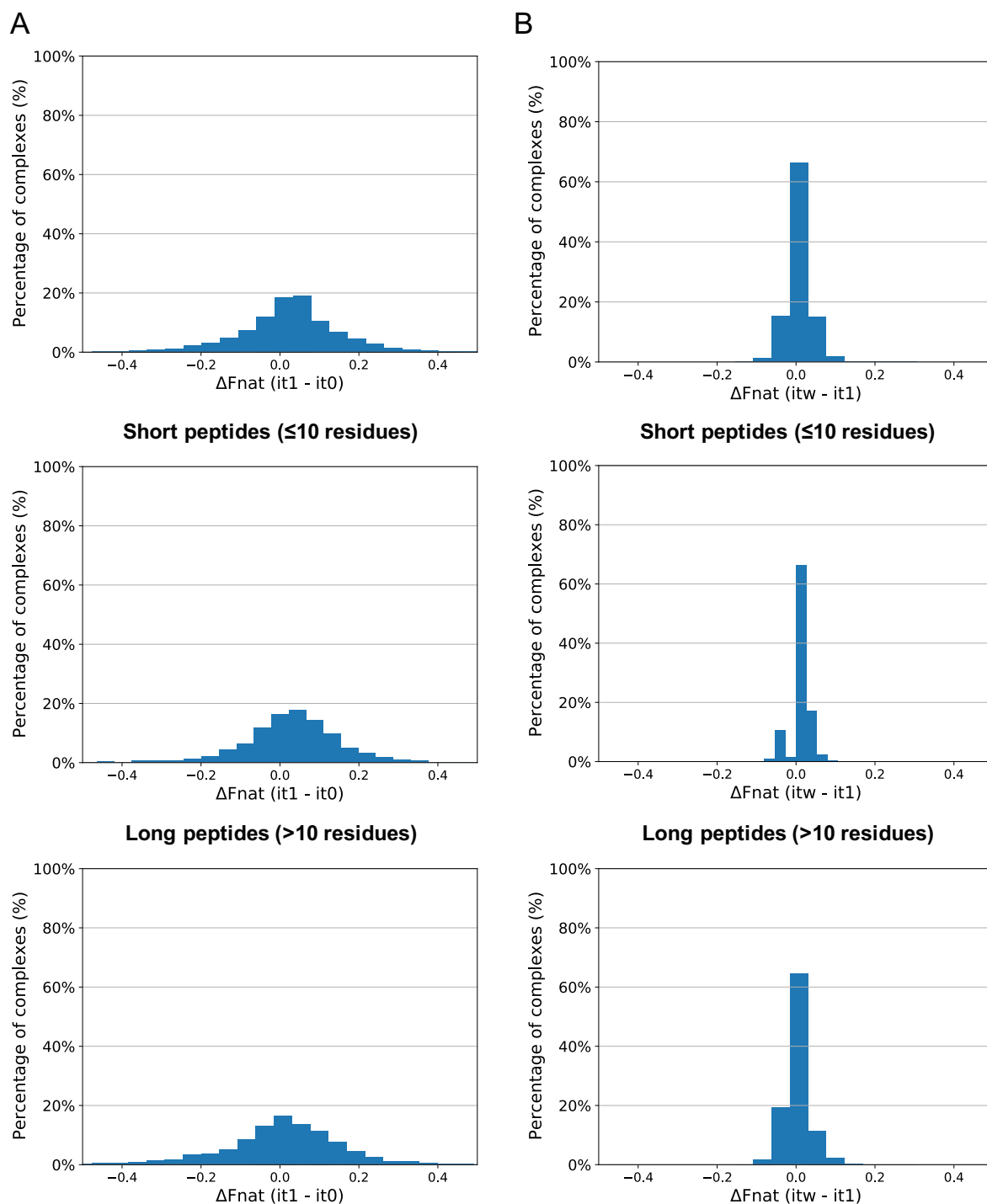


Figure S7. Difference in f_{nat} between models from various stages of HADDOCK (it0/it1/itw) for the 50STR_COMB experiment. The distributions are calculated from all generated models of the 50STR_COMB experiment using the *holo* structure of the receptor (30 complexes*400 models/complex = 12.000 data-points). **(A)** The impact of semi-flexible refinement in torsion angle space (it1-it0) is shown. Bin size used for the plot: 35 **(B)** The impact of final water refinement (or the simple energy minimization step, if short peptides are considered) is shown. Bin size used for the plot is 10.

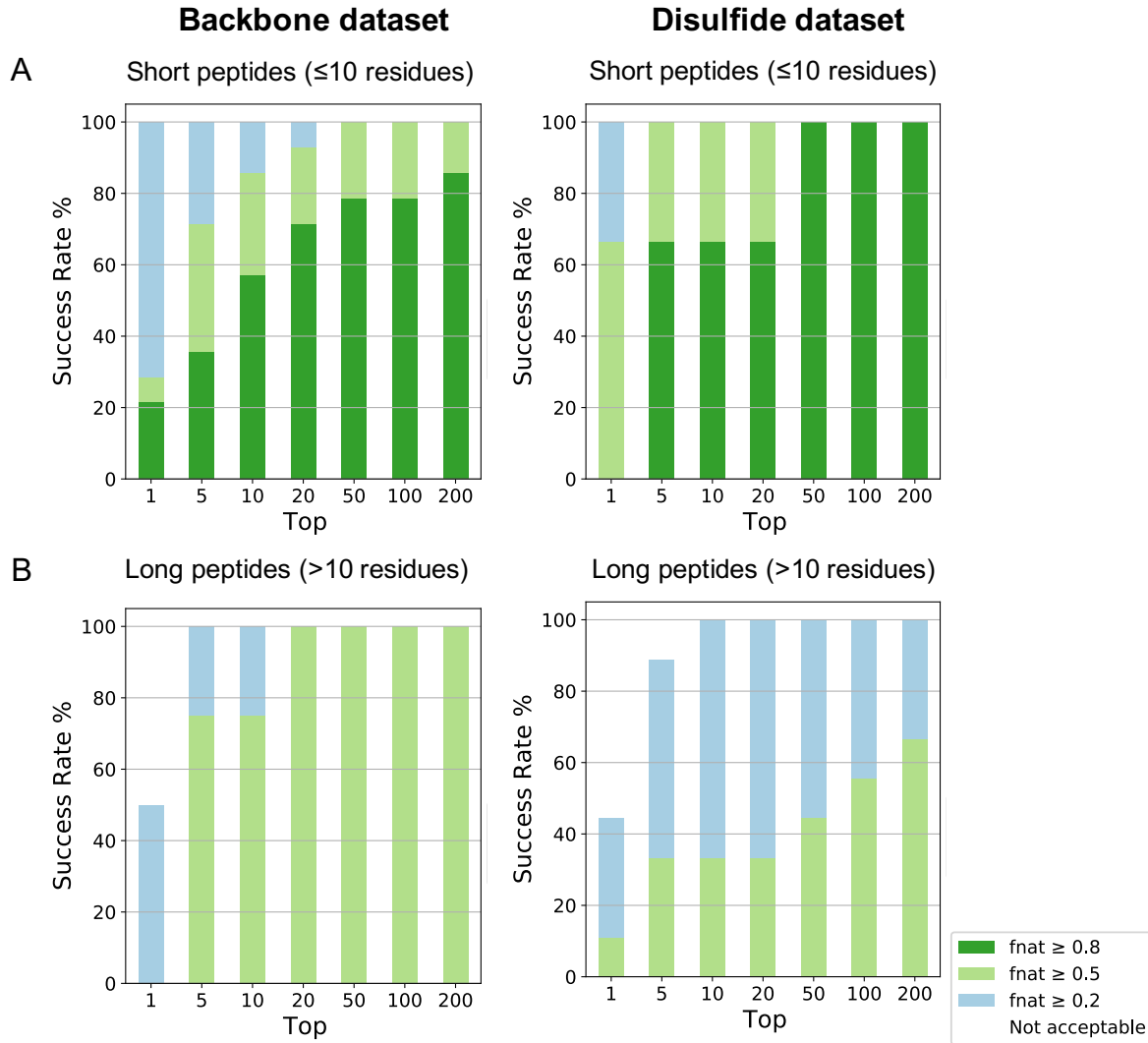


Figure S8. F_{nat} success-rate (%) plots of 50STR_COMB docking for short (≤ 10 residues) and long (> 10 residues) peptides separately in the Backbone and Disulfide dataset. (A) Plotted are the success-rates of the Backbone short peptides ($n=14$) and the Disulfide short peptides ($n=3$). (B) Plotted are the success-rates of the Backbone long peptides ($n=4$) and the Disulfide long peptides ($n=9$). Color coding (from blue to green) indicates the quality of models (from acceptable to high) according to CAPRI criteria.

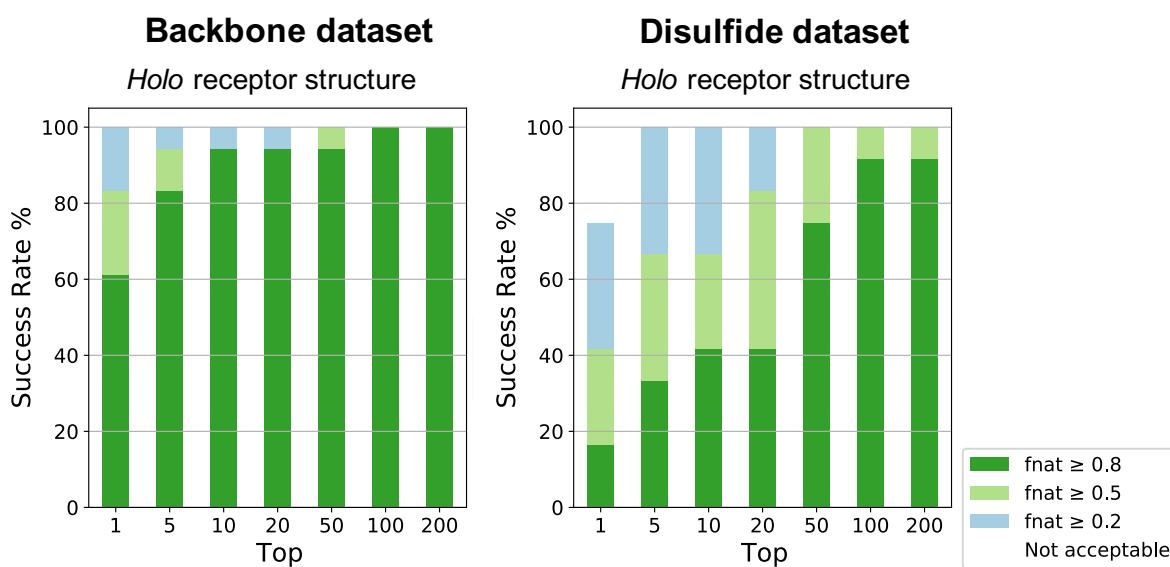


Figure S9. F_{nat} success-rate plots of best-case scenario (bound docking) for the Backbone and the Disulfide dataset. Plotted are the success-rates of docking the *holo* cyclic peptide structure with the *holo* receptor structure for both datasets. Color coding (from blue to green) indicates model quality (from acceptable to high), according to CAPRI criteria.

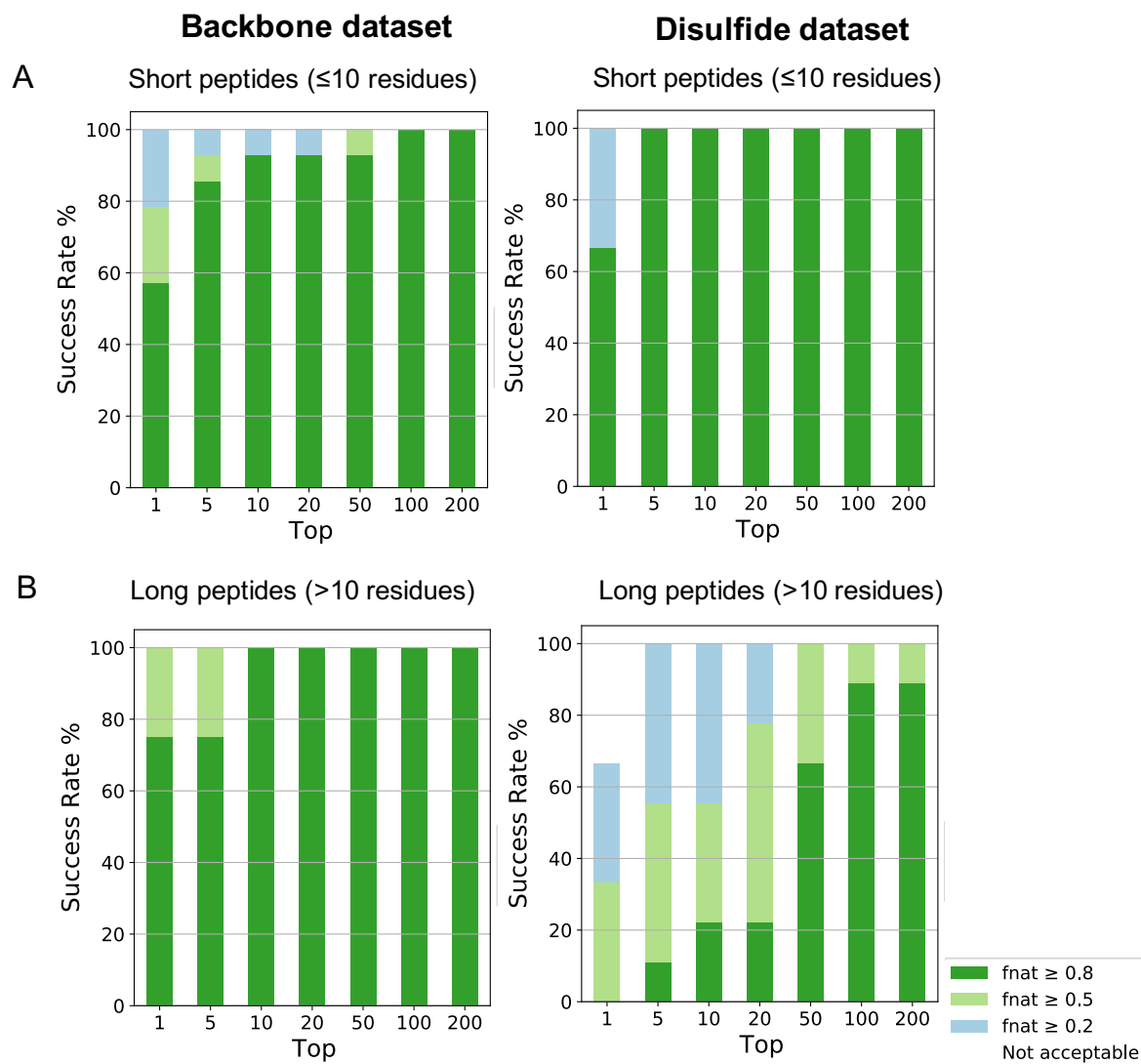


Figure S10. F_{nat} success-rate plots of best-case scenario (bound docking) for the Backbone and the Disulfide dataset. (A) Plotted are the success-rates of short peptides using the *holo* receptor structure (B) Plotted are the success rates of long peptides using the *holo* receptor structure. Color coding (from blue to green) indicates model quality (from acceptable to high), according to CAPRI criteria.

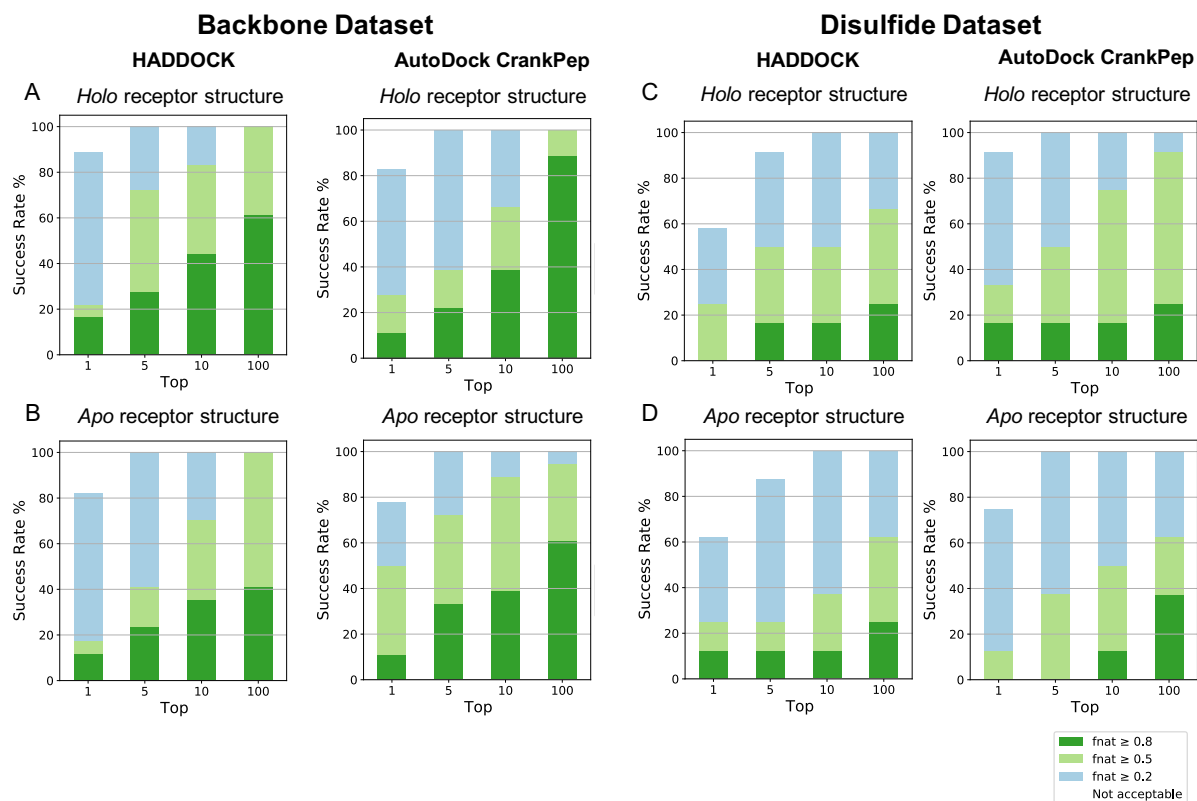


Figure S11. Comparison of HADDOCK's and AutoDock CrankPep's performance. (A) Plotted are the success rates of the optimized HADDOCK protocol for cyclic peptides (50STR_COMB) and the ADCP protocol for the Backbone dataset and the *holo* receptor (18 complexes) (B) or the *apo* receptor (17 complexes) (B) and for the Disulfide dataset with the *holo* receptor (12 complexes) (C) or the *apo* receptor (8 complexes) (D).